

Nguyen Duong Hieu

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Professional Summary

A dedicated Data Science student passionate about Artificial Intelligence, Large Language Models (LLMs), and Multimodal AI. Experienced in building end-to-end machine learning pipelines and integrating LLMs into enterprise workflows with a strong “build - test - iterate” mindset. Eager to leverage practical Python programming skills and a solid analytical foundation to develop and scale Agentic AI solutions in real-world business environments.

Education

National Economics University, BS in Data Science in Economics and Business Sept 2023 – Now

- GPA: 3.78/4.0
- **Coursework:** Linear Algebra, Probability & Statistics, Optimization, Data Preparation & Storytelling, Data Structures & Algorithms, Machine Learning, Database Systems (SQL).

Tran Phu High School for the Gifted, Natural Science 1 2020 – 2023

Experience

Trainee, AI Talent Training Program, Vingroup Summer 2026

- Selected for an intensive, hands-on training program focused on practical Artificial Intelligence applications.

Research Assistant, Project: “Does Network Position Cause Scholarly Success?” Spring 2026
with Prof. Mike Nguyen (University of Southern California) and Prof. Minh Nguyen (Florida Atlantic University)

- Curated, cleaned, and structured large-scale academic collaboration datasets, resolving inconsistencies across multiple sources to ensure data validity.
- Collaborated with the research team to support empirical and network-based analysis by providing high-quality, analysis-ready datasets.

Teaching Assistant (Tutor), Python for Data Science & Power Bi, National Economics University – Hanoi, Vietnam August 2025 – December 2025

- Supported students by explaining concepts in Python programming and data analysis, enhancing their practical skills in data science applications.
- Assisted the course instructor in preparing assignments and teaching materials for Python and Power BI courses.
- Evaluated and graded student submissions, providing detailed and constructive feedback to improve understanding.

Student Researcher, DATAOPT Lab – National Economics University, Hanoi, Vietnam July 2024 – January 2026

- Researched and implemented mathematical optimization algorithms (e.g., convex optimization) to strengthen foundations in applied mathematics.
- Evaluated the practical performance of numerical methods in machine learning contexts through experiment reproduction.

Projects

Multimodal AI in Enterprise Robotic Process Automation Spring 2026

- Designed a modular two-stage **agentic workflow framework** for multimodal information extraction and entity resolution to process unstructured business inputs.
- Integrated OCR and **ASR (PhoWhisper)** for data transcription, coupled with a **zero-shot LLM (Gemini)** for structured entity extraction.
- Developed a **hybrid retrieval mechanism** combining dense semantic representations (Sentence-BERT) with

sparse lexical matching (BM25) via Reciprocal Rank Fusion (RRF).

- Achieved F1-scores of 0.91 (visual) and 0.85 (acoustic) under noisy conditions, maintaining a Recall@5 of 0.93 on a 50,000-SKU enterprise catalog.

Hanoi Temperature Forecasting System [GitHub]

Late 2025

- Developed an end-to-end ML pipeline to forecast Hanoi's daily temperature (5-day horizon) using 10+ years of weather data.
- Engineered 90+ features (lag variables, rolling statistics, STL decomposition, FFT) to capture temporal patterns and seasonality.
- Trained a multi-output CatBoost model using Optuna for hyperparameter tuning, achieving $R^2 \approx 0.84$.
- Deployed an interactive Streamlit application integrated with SHAP for model explainability and an MLOps pipeline for data drift detection (PSI, KS-test).
- *Technologies: Python, Pandas, Scikit-learn, CatBoost, Optuna, SHAP, ONNX, Streamlit*

Skills

Technical Skills

- **Machine Learning & AI:** Large Language Models (LLMs), Prompt Engineering, API Integration, CatBoost, hyperparameter tuning (Optuna)
- **Programming:** Python (Pandas, NumPy, Matplotlib), SQL
- **Data Analysis:** Exploratory Data Analysis (EDA), statistical analysis
- **Mathematics:** Optimization, Probability & Statistics, Linear Algebra
- **Tools:** Power BI, Streamlit, LaTeX
- **Domain Knowledge:** Economics (microeconomics, macroeconomics fundamentals)

Publications

Promoting circular consumption behavior: Factors influencing Vietnamese Generation Z's intention to extend the lifespan of personal electronic devices

2026

Le Thu Thao, Nguyen Duong Hieu, Pham Phuong Trinh, Nguyen Minh Tung

22nd International Conference on Socio-economic and Environmental Issues in Development (ICSEED 2026)

- Investigated the determinants of Generation Z's intention to extend electronic device lifespans and the intention-behavior gap by integrating the Theory of Planned Behavior (TPB) and Push-Pull-Mooring (PPM) framework.
- Employed Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze survey data from 733 respondents.

Certifications and Achievements

- **2025** Top 10, Hanoi Mathematical Modeling Competition
- **2024** Academic Encouragement Scholarship (Excellent), Faculty of Mathematical Economics, National Economics University
- **2024** Outstanding Youth Union Member Award, National Economics University
- **2023** Consolation Prize, Hai Phong City Mathematics Competition
- **2022** IELTS 7.0

Interests

- Machine Learning, Large Language Models (LLMs), and Multimodal AI
- Agentic Workflows and Automation
- Optimization Methods for Machine Learning
- Applied Mathematics for Data-Driven Decision Making